Satvik Sharma

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Education

VIT Bhopal University — **September 2023 – May 2027**
B.Tech Computer Science and Engineering (AIML), CGPA: 8.21/10.0 — Sehore, MP

Saint Peter Senior Secondary School — **2023**
Class XII (10+2) — Chandigarh

Saint Soldier's Divine Public School — **2021**
Class X — Panchkula, Haryana

Skills

Programming & Web Technologies: C++, Python, JavaScript, HTML, CSS, React.js, RESTful APIs
Databases & Cloud platforms: SQL, MongoDB, AWS, Docker, Git, GitHub
AI/ML & MLOps: Scikit-learn, XGBoost, Pandas, Numpy, LangChain, RAG, DVC, MLflow, DagsHub
UI/UX Design & Productivity Tools: UI/UX Design, Figma, Microsoft Office (Excel, PowerPoint)

Experience

FOSSEE Semester Long Internship (IIT Bombay) — **February 2026 – June 2026**
Software Engineer Intern | **Completion Certificate: https://tinyurl.com/bdkky8pc** — Remote

- Delivered core full-stack features – spanning a React.js frontend and Node.js/Express.js backend – for a web-based, open-source electronics simulation and learning platform aiming to serve 1000+ concurrent users.
- Developed role-based dashboard systems, classroom management workflows, and a gamified progression system, increasing weekly user engagement by 30% and improving platform scalability.

Projects

SolarCarbon AI | *Python, FastAPI, Mendeley Data* | **Deployed Link: https://tinyurl.com/4e4er7tt**

- Renewable energy users had no reliable method to quantify **carbon credits** earned from solar panel usage, and accurately estimating grid emission factors demanded a physics-informed modeling approach capable of capturing non-linear relationships in raw energy data.
- Architected and trained a **Voting Regressor ensemble** combining **3 machine learning models** – HistGradientBoosting, RandomForest, and ExtraTrees – on a self-authored Mendeley dataset of **230+ thermal plant emission records**, and constructed a custom preprocessing pipeline integrating RobustScaler and PolynomialFeatures to model the data's non-linear structure.
- Deployed a production-ready prediction platform achieving an industry-competitive R^2 score of **0.97**, delivering accurate, real-time carbon credit estimation directly from user-submitted energy data.

OpenHW Studio | *React.js, Node.js, Express.js, MongoDB, TypeScript, Docker* | **Deployed Link: https://openhw-studio.fossee.in/**

- FOSSEE, IIT Bombay required a browser-based electronics simulation platform with no existing infrastructure, no mechanism to reduce instructor time lost to manual submission grading, and no secure data layer or onboarding flow for new users.
- Designed and built a full-stack platform across **4 areas** – frontend, backend, TypeScript emulator, and documentation – developing **twelve virtual hardware components** from scratch; engineered an intelligent assessment engine combining circuit-wiring analysis with regex-based C++ static analysis, gated at an **80% accuracy threshold**; and implemented **3 MongoDB schemas** – UserProgress, QuizAttempts, and CompletedProjects – with JWT-authenticated REST endpoints and a 14-step accessibility-compliant onboarding flow.
- Scaled the platform to support **1000+ concurrent users** simulating circuits without local hardware, eliminated **15+ instructor hours weekly** by streamlining grading through the assessment engine, and established a secure, fully onboarded data foundation the platform previously lacked.

RAG-Powered AI Code Reviewer | *ChromaDB, Sentence Transformers* | **Deployed Link: https://tinyurl.com/8m7daw9d**

- Tools like GitHub Copilot lacked an "institutional memory" of past code reviews, leaving pull request diffs unmatched against relevant precedent, while feedback still needed to arrive fast enough to avoid slowing merge velocity.
- Deployed a self-updating code review system as a **GitHub Webhook**; built a **FastAPI** backend to extract PR diffs via the GitHub API and retrieve relevant matches from a **ChromaDB** vector database powered by Sentence Transformers (`all-MiniLM-L6-v2`); and orchestrated a **LangChain** + **Groq** (`llama-3.1-8b`) **RAG** pipeline to generate structured feedback.
- Delivered context-aware, precedent-grounded, JSON-structured feedback within **5 seconds** of a Pull Request opening, matching the speed of manual first-pass review without blocking merge velocity.

Achievements

- **Hackathon Finalist:** Advanced to the Buildathon Final Review round out of 1000+ competing teams in **NASSCOM AI**, "Automation using Agentic AI" challenge, building an intelligent AI agent with Inya.ai for Consumer Durables.
- **Carbon Emission Dataset:** Published on Mendeley Data, adding and verifying latitude–longitude columns across 65,000+ records to support carbon credit research.

Certifications

- Applied ML in Python – University of Michigan, Coursera (2025)
- JLPT N5 - Japanese Language Proficiency Test (2025)
- Postman API – Postman (2024)